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B.Tech. DEGREE EXAMINATION, NOVEMBER 2012.

Third Semester

Computer Science and Engineering

DIGITAL SYSTEM DESIGN

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. What is the basic circuit for RTL and DTL logical families?
2. State De Morgan's law.
3. Draw the logic diagram for a 4-bit even parity checker.
4. Write the truth table for Full Subtractor.
5. Draw the logic diagram of SR Flip-flop.
6. List the differences between ripple counter and synchronous counter.

7. Write the steps taken for the purpose of transferring a new word to be stored in memory.
8. What are hazards?
9. Write the structure of VHDL program.
10. Define Constant.

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

#### UNIT I

11. Convert the following numbers to the indicated bases.
    - (a) Decimal 225 to binary, octal and hexadecimal. (6)
    - (b) Binary 11010111 to decimal, octal and hexadecimal. (5)
- Or
12. Simplify the following expression using K — map.
    - (a)  $F(x, y, z) = \Sigma(1, 2, 3, 6, 7)$  (4)
    - (b)  $F(x, y, z) = \Sigma(0, 1, 5, 7)$  (3)
    - (c)  $F(w, x, y, z) = \Sigma(2, 3, 12, 13, 14, 15)$ . (4)

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#### UNIT II

13. Implement BCD to Excess — 3 code. (11)
- Or
14. Explain in detail the 4 bit binary parallel adder. (11)

#### UNIT III

15. Explain in detail the edge triggered flip-flop. (11)
- Or
16. Explain with neat diagram the serial adder using shift register. (11)

#### UNIT IV

17. Explain with a neat diagram the PAL. (11)
- Or
18. Explain in detail the error correcting code. (11)

#### UNIT V

19. Explain in detail the functions and procedures of VHDL. (11)
- Or
20. Explain in detail the VHDL for combinational circuit. (11)

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B.Tech. DEGREE EXAMINATION, NOVEMBER 2013.

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DIGITAL SYSTEM DESIGN

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Expand RTL, DTL, TTL, ECL.
2. Convert the following decimal to binary  
(a)  $(0.6875)_{10}$ .  
(b)  $(41)_{10}$ .
3. Differentiate between encoder and decoder.
4. Define Parity bit.
5. Give the characteristics table of JK Flip-Flop
6. What is the use of registers?

7. Define memory unit.
8. Give the types of ROM.
9. What is VHDL?

10. Describe packages in VHDL.

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

#### UNIT I

11. Determine the prime — implicants of the function.  
 $F(x, y, z) = \Sigma(1, 4, 6, 7, 8, 9, 10, 11, 15).$

Or

12. (a) Write short notes on Minterms and Maxterms. (5)
- (b) Simplify Boolean function.  
 $F(A, B, C, D, E) = \Sigma(0, 2, 4, 6, 9, 11, 13, 15, 17, 21, 25, 27, 29, 31).$  (6)

#### UNIT II

13. Write in detail about Half Adder and Full Adder with logic diagram.

Or

14. Write about Multiplexer and Demultiplexer.

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#### UNIT III

15. Describe clocked sequential circuit.

Or

16. Explain about Synchronous Counters.

#### UNIT IV

17. Explain error detection and correction in detail.

Or

18. Describe in detail RAM with neat block diagram.

#### UNIT V

19. Explain the functions and procedures of VHDL in detail.

Or

20. Describe the model of a generic N by M binary multiplier.

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UNIT V

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19. Explain about the richness of operators available in VHDL. (11)

Or

20. Explain the following statements available in VHDL.
- |                       |     |
|-----------------------|-----|
| (a) Process statement | (5) |
| (b) Loop statement.   | (6) |

B.Tech. DEGREE EXAMINATION, APRIL/MAY 2014.

Third Semester

Computer Science and Engineering  
DIGITAL SYSTEM DESIGN

Time : Three hours                      Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Convert (673.124)<sub>10</sub> into a binary number.
2. State de Morgan's law.
3. Give the characteristics of CMOS family.
4. Compare totem pole and open collector output of TTL family.
5. Implement  $f = abcd$  using two input NAND Gate.
6. What is the difference between multiplexer and demultiplexer?
7. Define race around condition and explain how it can be eliminated.

8. Define lockout condition.

9. What are the types of asynchronous circuit?

10. Define essential hazard.

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each unit.

All questions carry equal marks.

#### UNIT I

11. Draw the circuit diagram and explain the operation of two TTL NAND gate with open collector output. (11)

Or

12. Simplify the following Boolean function by using the tabulation method  
 $F(w, x, y, z) = \Sigma(2, 3, 12, 13, 14, 15)$ . (11)

#### UNIT II

13. Design a binary to gray code converter. (11)

Or

14. (a) What is a decoder? Give the truth table of 3 : 8 decoder and supplement it using basic gate. (6)  
(b) Design a 4 : 1 Mux. (5)

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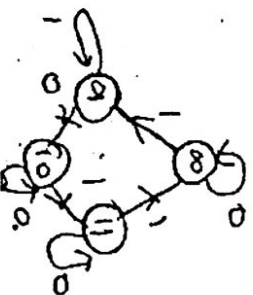
#### UNIT III

15. Design a synchronous counter for  
 $4 \rightarrow 6 \rightarrow 7 \rightarrow 3 \rightarrow 1 \rightarrow 4 \dots$

Avoid lock out condition. Use JK type design. (11)

Or

16. Design a sequential circuit for the given state diagram. (11)



#### UNIT IV

17. Write notes on Asynchronous and synchronous sequential circuits design. (11)

Or

18. Design a gated latch with two inputs G(gate) D(data) and one output Q. (11)

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UNIT V

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19. Describe the VHDL for combinational circuits in detail.

Or

20. Explain the VHDL for binary multiplier in detail.

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NOVEMBER/DECEMBER 2014.

Third Semester

Computer Science and Engineering

DIGITAL SYSTEM DESIGN

Time : Three hours

Maximum : 75 marks

SECTION A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Convert the binary number 0.0110 into decimal.
2. Simplify the given Boolean functions to a minimum number of literals  $(x + y)(x + y')$ .
3. Define: Decoder.
4. Write short note on Multiplexers.
5. Discuss briefly about D-latch.
6. List the features of shift registers.

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7. Write the different types of ROM.

8. What is meant by race-free state?

9. Discuss the program structure of VHDL?

10. Write down the different data types in VHDL.

SECTION B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

#### UNIT I

11. Draw the logic diagram corresponding to the following Boolean expressions without simplifying them.

(a)  $BC' + AB + ACD$

(b)  $(A + B)(C + D)(A' + B + D)$

(c)  $(AB + A'B')(CD' + C'D)$

Or

12. Express the given function in sum of min-terms and product of max-terms

$$F(A, B, C, D) = B'D + A'D + BD$$

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#### UNIT II

13. Describe the 4 bit by 3 bit binary multiplier in detail.

Or

14. Describe the logic diagram of carry look ahead generator in detail.

#### UNIT III

15. Describe in detail the analysis of clocked sequential circuits with D and JK flip-flops.

Or

16. Describe the shift registers and ring counter in detail.

#### UNIT IV

17. Explain in detail about the programmable logic array with 3 inputs, 2 outputs and 4 product terms.

Or

18. Discuss the design procedure of asynchronous sequential circuit in detail.

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UNIT IV

17. Explain programmable logic arrays. (PLAs)

Or

18. Explain programmable logic devices. (PLDs)

UNIT V

19. Explain the modeling styles for combinational logic.

Or

20. Implement counter design using VERILOG HDL.

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B.Tech. DEGREE EXAMINATION,  
APRIL 2015.

Third Semester

Computer Science and Engineering  
DIGITAL SYSTEM DESIGN  
(2013-2014 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Convert to the decimal number 250.5 to base 8 and base 7.
2. Convert  $F = xy + x'z$  to product of maxterms.
3. How many don't-case inputs are there in a BCD adder?
4. Write the truth table for full subtractor.
5. Draw the logic diagram of Master slave flip-flop.
6. Write excitation table for JK flip-flop.
7. Differentiate ROM from PLA.

8. Do's and Don'ts of FPGA's.
9. Define Behavioral models.
10. What are the datatypes allowed in VERILOG.HDL?

PART B --- (5 × 11 = 55 marks)

Answer ALL questions one from each unit.

All questions carry equal marks.

#### UNIT I

11. Perform subtraction with the following decimal numbers using 10's complement and 9's complement.
  - (a) 5250-321
  - (b) 3570-2100
  - (c) 753-864
  - (d) 20-1000.

Or

12. Simplify the following boolean function using Quine-Mcclusky method.

$$F = \Sigma(0,2,4,6,8,9,10,11,12,14)$$

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#### UNIT II

13. Explain the working principle of half adder and full adder.

Or

14. Design a combinational circuit with a decoder and external gates using the following function.

$$F_1 = x'y' + xy'z'$$

$$F_2 = x' + y$$

$$F_3 = xy + x'y'$$

#### UNIT III

15. Design a counter with the following binary sequence: 0,1,3,2,6,4,5,7 and repeat. Use RS flip-flop.

Or

16. A sequential circuit has 2 flip-flop, 2 inputs and an output. The flip-flop input function and circuit output function are as follows:

$$JA = xB + y'B'$$

$$KA = xy'B'$$

$$JB = xA'$$

$$KB = xy'A + A$$

$$Z = xy'A + x'y'B$$

Obtain the logic diagram, state table, state diagram and state equation.

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#### UNIT IV

17. (a) A combinational circuit is defined by the functions:  $F1 = \sum(1,3,5)$  and  $F2 = \sum(5,6,7)$  implement the circuit with a PLA having 3 inputs and 3 product terms (7)  
 (b) Explain the steps involved in programming of FPGA. (4)

Or

18. Implement the following using PLA.  
 $A(x,y,z) = \sum(0,1,2,5)$   
 $B(x,y,z) = \sum(0,3,5,7)$   
 $C(x,y,z) = \sum(0,1,2,3).$

#### UNIT V

19. (a) Write the HDL behavioral description of a 4 to 1 line Multiplexer.  
 (b) Implement full adder circuit using VHDL
- Or
20. (a) Realize 1X8 DEMUX using VHDL. (6)  
 (b) Explain the behavioral approach for hardware description with an example (5)

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B.Tech. DEGREE EXAMINATION, APRIL/MAY 2016.

Third Semester

Computer Science and Engineering

DIGITAL SYSTEM DESIGN

Time : Three hours

Maximum : 76 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

1. State DeMorgan's law and prove it.
2. Express the function  $F = (xy + z)(y + xz)$  in SOP form.
3. Draw the logic diagram of 2 to 4 line decoder using NAND gates.
4. What do you know about a magnitude comparator? How many outputs are there in the circuit?
5. A PN flip-flop has four operations: clear to 0, no change, complement, and set to 1, when inputs P and N are 00,01,10 and 11 respectively. Tabulate the characteristic table.

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6. What do you know about a ring counter? Draw a four bit ring counter.
7. What is the purpose of using decoder in a memory chip?
8. A memory stores  $8K \times 16$  bytes words. How many address lines and data input / output does it have?

9. State the purpose of using HDL modeling.

10. Write a HDL program for Half Subtractor.

PART B — ( $5 \times 11 = 55$  marks)

Answer ALL questions, ONE from each Unit.

#### UNIT I

11. Simplify the Boolean function using Quine — McCluskey method :

$$F = \Sigma(0, 2, 3, 5, 7, 8, 9, 10, 11, 13, 15).$$

Or

12. Draw the NAND logic diagram that implements the following function after simplified using K map :  $F(A, B, C, D) = \Sigma(0, 1, 2, 3, 4, 8, 9, 12).$

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#### UNIT II

13. Explain in detail about BCD adder circuit and design a 4 bit BCD adder.

Or

14. (a) Implement the Boolean function  $F(A, B, C, D) = \Sigma(0, 1, 2, 5, 7, 11, 14)$  with multiplexer. (5)
- (b) Discuss in detail about parity encoder. Design a parity encoder with 4 bit input. (6)

#### UNIT III

15. (a) Design a synchronous counter that goes through the sequence 2, 6, 1, 7, 5, 4 and repeat. (7)

(b) Draw the logic diagram of a 4-bit universal shift register with four D flipflop and four  $4 \times 1$  mux with mode selection inputs S1 and S0 and explain it. (4)

Or

16. (a) Design a JK flip flop using D flip flops. (4)
- (b) Design a sequential circuit with two T flip-flops A and B and two inputs E and x. if  $E = 0$ , the circuit remains in the same state regardless of the value of x. When  $E = 1$  and  $x = 1$ , the circuit goes through the state transitions from 00 to 01 to 10 to 11 back to 00, and repeats. When  $E = 1$  and  $x = 0$ , circuit goes through the state transitions from 00 to 11 to 10 to 01 back to 00, and repeats. (7)

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B.Tech. DEGREE EXAMINATION, APRIL/MAY 2016.

Third Semester

Computer Science and Engineering

DIGITAL SYSTEM DESIGN

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Convert the 364 decimal into BCD equivalent.
2. State error detecting and correction codes.
3. Convert Gray number 1110 to its BCD equivalent.
4. Annotate encoder.
5. Define counters.
6. Annotate the shift register
7. List out the types of memories.



8. Differentiate between PLA and PAL.
9. Define modules in verilog.
10. What is meant by full adder?

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each unit.

All questions carry equal marks.

#### UNIT I

11. Discuss the boolean algebra and theorem with example.

Or

12. Elucidate the minimization of expression using Karnaugh maps and Quine-McCluskey.

#### UNIT II

13. Explain half adder and full adder with a neat circuit diagram.

Or

14. Illustrate multiplexer and de-multiplexer with a neat circuit diagram.

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#### UNIT III

15. Explain the edge-triggered JK and D flip flops.

Or

16. Design GCD processor with necessary state table and state reduction techniques.

#### UNIT IV

17. Explain how the RAM and ROM chips can be organized and how the addresses are decoded to select those chips.

Or

18. Illustrate the combinational logic implementations using PROMs, PLAs, PALs.

#### UNIT V

19. Elucidate the 4-bit ripple carry counter with a neat diagram.

Or

20. Discuss the data types used in Verilog.

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B.Tech. DEGREE EXAMINATION,  
NOVEMBER/DECEMBER 2016.

Third Semester

Computer Science and Engineering

DIGITAL SYSTEM DESIGN

(2013-2014 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Find the octal equivalent of decimal 200.
2. State De Morgan's theorem.
3. What is meant by decoder?
4. What is multiplexer?
5. What is ring counter?
6. What is JK flip-flop?
7. Distinguish between RAM and ROM.

8. What is meant by address decoding?
9. What is meant by gate-level modeling?
10. What is meant by dataflow modeling?

PART B — (5 × 11 = 55 marks)

Answer ALL questions, choosing ONE from each Unit.  
All questions carry equal marks.

#### UNIT I

11. Using the K-map method, obtain the minimal sum of product expression of the following function  
 $y = \Sigma(0, 2, 3, 6, 7, 8, 10, 11, 12, 15)$ .

Or

12. Find the minimal sum of products for the Boolean expression  
 $f(w, x, y, z) = \Sigma(1, 3, 4, 5, 9, 10, 11) + \Sigma d(6, 8)$ .  
Using the Quine-McClusky method. And also draw the logic circuit diagram.

#### UNIT II

13. Design a full subtractor using only NOR gates.

Or

14. Briefly discuss about parallel adders.  
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#### UNIT III

15. Explain about different types of Shift Registers in detail.

Or

16. Discuss about mealy and Moore models in detail.

#### UNIT IV

17. Briefly discuss about Programmable Array Logic (PAL) devices.

Or

18. Explain about different types of memories in detail.

#### UNIT V

19. Write notes on hierarchical Modeling concepts.

Or

20. Explain about 4-bit ripple carry counter in detail.

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**RAJIV GANDHI COLLEGE OF ENGINEERING AND TECHNOLOGY**  
**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**  
**CS T34 – DIGITAL SYSTEM DESIGN- INTERNAL TEST-II**

**YEAR/SEM/SEC: Y2/S3/A&B**

**MAX. MARKS: 40**

**DATE: 27-08-2015**

**TIME: 1.30 Hrs**

**PART-A**

**(5X2=10)**

**Answer all the Questions:**

1. What is the difference between multiplexer and demultiplexer?
2. Implement  $f = abcd$  using two input NAND Gate.
3. Distinguish between Synchronous and Asynchronous sequential circuit?
4. Define counter.
5. What is meant by clocked sequential circuit?

**PART-B**

**(3X10=30)**

6. Explain Encoder and Decoder in detail.  
(OR)
7. Discuss Multiplexer and Demultiplexer.
8. (a) Describe Master-Slave flip flop. (6)  
(b) Describe JK flip flop. (4)  
(OR)
9. Explain in detail about SR Latch.
10. Explain Shift Register.  
(OR)
11. Describe the working principle of Binary Ripple counter.

**RAJIV GANDHI COLLEGE OF ENGINEERING AND TECHNOLOGY**  
**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**  
**CS T34 – DIGITAL SYSTEM DESIGN- INTERNAL TEST-II**

**YEAR/SEM/SEC: Y2/S3/A&B**

**MAX. MARKS: 40**

**DATE: 27-08-2015**

**TIME: 1.30 Hrs**

**PART-A**

**(5X2=10)**

**Answer all the Questions:**

6. What is the difference between multiplexer and demultiplexer?
7. Implement  $f = abcd$  using two input NAND Gate.
8. Distinguish between Synchronous and Asynchronous sequential circuit?
9. Define counter.
10. What is meant by clocked sequential circuit?

**PART-B**

**(3X10=30)**

12. Explain Encoder and Decoder in detail.  
(OR)
13. Discuss Multiplexer and Demultiplexer.
14. (a) Describe Master-Slave flip flop. (6)  
(b) Describe JK flip flop. (4)  
(OR)
15. Explain in detail about SR Latch.
16. Explain Shift Register.  
(OR)
17. Describe the working principle of Binary Ripple counter.

**RAJIV GANDHI COLLEGE OF ENGINEERING AND TECHNOLOGY**  
**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**  
**MODEL EXAMINATION (OCT-2017)**

**SUBJECT NAME/CODE: DIGITAL SYSTEM DESIGN (CST 34)**

**YEAR /SEM: II/S3/A&B**

**DATE: 25.10.17**

**ANSWER ALL QUESTIONS:**

**25 X 2 = 50 MARKS**

1. Define digital system design.
2. Convert the binary number 0.0110 into decimal.
3. Convert (475.12) base 8 into binary number.
4. Define gray code
5. State de Morgan's law.
6. What is the difference between multiplexer and demultiplexer.
7. Define decoder.
8. Discuss briefly about D-Latch
9. List the features of shift registers.
10. Give the Boolean expression for full-adder.
11. State the distributive law of Boolean algebra.
12. Define sequential circuit and list the types
13. What is the use of register?
14. Define memory unit
15. Define k-map with 3 variable representations.
16. Write down the different data types in VHDL.
17. Define ROM and list the types.
18. State mealy and moore model.
19. What are the types of asynchronous circuit?
20. Give the characteristics table of JK Flip Flop.
21. What are the types of asynchronous circuit?
22. What is the difference between PLA and PAL?
23. Difference between sequential and combinational circuit.
24. Define parity bit
25. What is VHDL?

60/50